Dt. 24/09/2022

AUTUMN MID-SEMESTER EXAMINATION- 2022-23

School of Electronics Engineering
Kalinga Institute of Industrial Technology
Deemed to be University

5th Semester

Subject: ARM and Advanced Processors [EC 3031]

Time: 1½ Hours Full Marks: 20

The figures in right-hand side indicates full marks.
All parts of a question should be answered at one place only.

Question No.	Section -A	Question	CO Mapping	Marks
Q1	SAT			[1x5]
a)		Explain the role of Task Register and Task State Segment during multi-tasking? Compute the maximum physical address space in REAL mode and PVAM mode that 80286 can access.	CO1	
b)		DS segment selector presently contains 000FH.Determine whether it would access GDT or LDT? If the DPL bits of the accessed descriptor is 11 (level-3),will the access be granted?	CO2	
c)		What is Page Fault Linear Address? Where is it stored in a 80386processor?	CO3	
d)		What does Granularity (G) bit of 80386 specify?	CO3	
e)		How many descriptors are stored in GDT of 80386 if the limit value stored at GDTR register is 032FH and the base address is 04200000H?	CO3	
	Section - B			
Q2	What is Multi-user or Multi-tasking concept in higher level Processors? Identify the need for memory management in higher level processors in multi-user/ multi-tasking environment?		CO1	[5]
	OR			
	Examine the procedure of memory management in higher level processors.		CO1	[5]
Q3	Examine how Logical address is converted to Physical address in a 80286 processor assuming TI bit in the selector as 1.		CO2	[5]
	OR			
	What is the purpose of a Shadow Registers in a 80286 processor? Explain the functions of PEREQ PEACK, BUSY and ERROR pins of 80286 processor.		CO2	[5]
Q4	Discuss the address translation mechanism with paging option 'ON' to convert a Linear address to Physical address in a 80386 system.		CO3	[5]
	OR			
	Discuss the concept of memory banks in 80386 microprocessor. If the status of byte enable lines of 80386 $\overline{BE3}, \overline{BE2}, \overline{BE1}, \overline{BE0}$ are 1101, then 8bit data will flow through which part of the data bus?		CO3	[5]